Zhifan Zhou

Curriculum Vitae

+1-240-521-9320
zhifanzhou12@gmail.com
https://zhifanzhou.com

Current Appointment

Nov. 2019-present — Postdoctoral Researcher, **Institute for Physical Science and Technology (IPST) and Joint Quantum Institute (JQI), University of Maryland and National Institute of Standards and Technology (NIST)**, USA.
-Advisor, Avik Dutt & Paul Lett

Education

2008-2013 — **Ph.D. in Optics**, *East China Normal University*, Shanghai, China.
including an exchange program from 2010 to 2011 at JQI in NIST and University of Maryland, USA.
- Thesis title: Manipulation of Photons based on Coherent Atomic Ensemble.

2004-2008 — **B.Eng. in Electronic Science and Technology**, *East China Normal University*, Shanghai, China.

Research Interests

I am interested in various topics related to quantum science and atomic, molecular, and optical physics, including quantum sensing, quantum information processing, ultracold atoms, and precision measurement.

Research Experience

2014-2019 — Postdoctoral Researcher, **Ben-Gurion University of the Negev**, Israel.
- Advisor: Ron Folman

2008-2011 & 2011-2013 — PhD Student/Research Assistant, **East China Normal University**, China.
- Advisor: Jietai Jing & Weiping Zhang

2010-2011 — Visiting PhD Student, **National Institute of Standards and Technology**, USA.
- Advisor: Paul Lett

Academic Activities

Session chair: 54th Annual Meeting (2023) of the APS (American Physical Society) DAMOP (Division of Atomic, Molecular and Optical Physics) (Session E10: Open Quantum Systems),
Quantum Sensing, Imaging, and Precision Metrology II (2024) of (SPIE (Society of Photo-Optical Instrumentation Engineers) Photonics West 2024) (Session 8: Integrated Photonics, Nano Photonics, and Microphotonics I).
Journal reviewer: Physical Review Letters, Optica Quantum, Physical Review Applied, Physical Review A, Photonics Research, Optics Letters, Optics Express, and Journal of the Optical Society of America A and B.

Publications

Summary — 20+ publications with 1000+ citations and an h-index of 14 (according to Google Scholar), including 1 paper in Science, 3 papers in Science Advances, 1 paper in Physics Review Letters, 1 paper in Optica Quantum, 3 papers in Applied Physics Letters, 1 paper in Classical and Quantum Gravity, 2 papers in New Journal of Physics, 2 papers in Optics Letters, and 4 papers in Optics Express.

Preprints

2024 [P1] **Generation of hypercubic cluster states in 1-4 dimensions in a simple optical system**.
Zhifan Zhou, Luís E. E. de Araujo, Matt DiMario, Jie Zhao, Jing Su, B. E. Anderson, Kevin M. Jones, Paul D. Lett
arXiv:2408.06317, 2024. In revision for Science Advances.

Published Journals

2024 [J20] **Enhanced metrology via geometric phase amplification in a clock interferometer**.
Zhifan Zhou, Sebastian C. Carrasco, Christian Sanner, Vladimir S. Malinovsky, and Ron Folman
Science Advances, 11, eadr6893 (2025).

2024 [J19] **Color-switching in an optical parametric oscillator using a phase-conjugate mirror**.
B. E. Anderson, Jie Zhao, **Zhifan Zhou**, R. Speirs, Kevin M. Jones, Paul D. Lett
Optics Express, 32, 23812-23821 (2024).

2024 [J18] **Characterizing two-mode-squeezed light from four-wave mixing in rubidium vapor for quantum sensing and information processing**.
Luís E. E. de Araujo, **Zhifan Zhou**, Matt DiMario, B. E. Anderson, Jie Zhao, Kevin M. Jones, Paul D. Lett
Optics Express, 32, 1305-1313 (2024).

2023 [J17] **Nonlocal phase modulation of multimode, continuous-variable twin beams**.
Zhifan Zhou, Luís E. E. de Araujo, Matt DiMario, B. E. Anderson, Jie Zhao, Kevin M. Jones, and Paul D. Lett
Optica Quantum, 1, 71-77 (2023).

2022 [J16] **Observation of anomalous Moire patterns**.
Omer Amit, Or Dobkowski, **Zhifan Zhou**, Yair Margalit, Yonathan Japha, Samuel Moukouri, Yigal Meir, Baruch Horovitz, Ron Folman
New Journal of Physics, 24, 073032 (2022).

2021 [J15] **Realization of a complete Stern-Gerlach interferometer: Towards a test of quantum gravity**.
Yair Margalit, Or Dobkowski, **Zhifan Zhou**, Omer Amit, Yonathan Japha, Samuel Moukouri, Daniel Rohrlich, Anupam Mazumdar, Sougato Bose, Carsten Henkel, Ron Folman.
Science Advances, 7, eabg2879 (2021).

2020 [J14] **An experimental test of the geodesic rule proposition for the non-cyclic geometric phase**.
Zhifan Zhou, Yair Margalit, Samuel Moukouri, Yigal Meir, Ron Folman.
Science Advances, 6, eaay8345 (2020).

2019 [J13] **Analysis of a high-stability Stern-Gerlach spatial fringe interferometer**.
Yair Margalit, **Zhifan Zhou**, Shimon Machluf, Yonathan Japha, Samuel Moukouri, Ron Folman.
New Journal of Physics, 21, 073040 (2019).

[J12] **T^3-Stern-Gerlach Matter-Wave Interferometer**.
O. Amit, Y. Margalit, O. Dobkowski, **Z. Zhou**, Y. Japha, M. Zimmermann, M. A. Efremov, F. A. Narducci, E. M. Rasel, W. P. Schleich, R. Folman.
Physics Review Letters, 123, 083601 (2019).

2018 [J11] **Quantum complementarity of clocks in the context of general relativity**.
Zhifan Zhou, Yair Margalit, Daniel Rohrlich, Yonathan Japha, Ron Folman.
Classical and Quantum Gravity, 35, 185003 (2018).

2015 [J10] **A self-interfering clock as a "which path" witness**.
Yair Margalit, **Zhifan Zhou**, Shimon Machluf, Daniel Rohrlich, Yonathan Japha, Ron Folman.
Science, 349, 1205 (2015).

2014 [J09] **Ultralow-light-level all-optical transistor in rubidium vapor**.
Jietai Jing, **Zhifan Zhou**, Cunjin Liu, Zhongzhong Qin, Yami Fang, Jun Zhou, Weiping Zhang.
Applied Physics Letters 104, 151103 (2014).

2012 [J08] **Optical logic gates using coherent feedback**.
Zhifan Zhou, Cunjin Liu, Yami Fang, Jun Zhou, Ryan T. Glasser, Liqing Chen, Jietai Jing, Weiping Zhang.
Applied Physics Letters 101, 191113 (2012).

[J07] **Temporally multiplexed storage of images in a gradient echo memory**.
Quentin Glorieux, Jeremy B. Clark, Alberto M. Marino, **Zhifan Zhou**, Paul D. Lett.
Optics Express, 20, 12350 (2012).

[J06] **Storing a short movie in an atomic vapor**.
Quentin Glorieux, Jeremy B. Clark, Alberto M. Marino, **Zhifan Zhou**, Paul D. Lett.
SPIE Newsroom, DOI: 10.1117/ 2.1201206.004337(2012).

[J05] **Imaging using quantum noise properties of light**.
Jeremy B. Clark, **Zhifan Zhou**, Quentin Glorieux, Alberto M. Marino, Paul D. Lett.
Optics Express, 20, 17050 (2012).

[J04] **Compact diode-laser-pumped quantum light source based on four-wave mixing in hot rubidium vapor**.
Zhongzhong Qin, Jietai Jing, Jun Zhou, Cunjin Liu, Raphael Pooser, **Zhifan Zhou**, Weiping Zhang
Optics Letters, 37, 3141 (2012).

[J03] **Squeezing bandwidth controllable twin beam light and phase sensitive nonlinear interferometer based on atomic ensembles**.
Jietai Jing, Cunjin Liu, **Zhifan Zhou**, et al.
Chinese Science Bulletin, 57, 1925 (2012).

2011 [J02] **Realization of a nonlinear interferometer with parametric amplifiers**.
Jietai Jing, Cunjin Liu, **Zhifan Zhou**, Z. Y. Ou, Weiping Zhang.
Applied Physics Letters 99, 011110 (2011).

[J01] **Realization of low frequency and controllable bandwidth squeezing based on a four-wave-mixing amplifier in rubidium vapor**.
Cunjin Liu, Jietai Jing, **Zhifan Zhou**, R. C. Pooser, F. Hudelist, Lu Zhou, Weiping Zhang.
Optics Letters, 36, 2979 (2011).

Talks and Presentations

Invited Talks

01/2024 **Enhanced metrology with the geometric phase jump in a clock interferometer**. Session: Recent Advances in Quantum Sensing, Quantum Sensing, Imaging, and Precision Metrology **SPIE. Photonics West 2024.**

01/2024 **Quantum correlated four-wave-mixing: Creation of n-dimensional optical cluster states**. Session: Entangled States of Photons and Non-Hermitian Sensors, Quantum Sensing, Imaging, and Precision Metrology **SPIE. Photonics West 2024.**

01/2024 **Enhanced metrology with the geometric phase jump in a clock interferometer**. Session: Information processing and sensing with ultracold atoms, the 53rd Winter Colloquium on the Physics of Quantum Electronics **PQE-2024.**

11/2023 **Generation of hypercubic cluster states in 1-4 dimensions in a simple optical system**. **Joint Quantum Institute (JQI) Postdoc Seminar, 2023.**

01/2023 **Nonlocal phase modulation of continuous-variable twin beams**. Quantum Sensing, Imaging, and Precision Metrology **SPIE. Photonics West 2023.**

01/2020 **An experimental test of the geodesic rule proposition for the noncyclic geometric phase**. Session: Atom Optics and Interferometry II, the 50th Winter Colloquium on the Physics of Quantum Electronics **PQE-2020.**

08/2015 **Coherent Stern-Gerlach matter-wave interferometer on an atom chip**. the 3rd International Conference on Molecular, Atomic, Nuclear and Particle Physics, Shanghai, China **MANPP 2015.**

Oral Presentations at Conferences

06/2023 **Internal clock interferometry for testing fundamental physics**. **DAMOP 2023**. C05.9

10/2022 **Nonlocal phase modulation of continuous-variable twin beams**. **FiO+LS 2022**. FW1B.2

05/2022 **Dissipative Kerr solitons in a warm atomic vapor system**. **CLEO 2022**. FS4B.8

06/2020 **Freezing of spinor dynamics in an ultracold Bose gas via microwave dressing**(virtual). Annual Meeting of the APS **DAMOP 2020**. N09. 00001

12/2018 **Quantum complementarity with clocks in the context of general relativity**. The Israel Physical Society annual meeting 2018, the section of Atomic, Molecular and Optical B, Jerusalem, Israel **2018 IPS**.

03/2014 **Detection of single photons using slow light**. Quantum Information and Measurement **QIM 2014** Berlin, Germany, paper: QTu2B.5.

06/2011 **Dual all-optical OR/NOR logic gates in hot rubidium vapor**. Annual Meeting of the APS **DAMOP 2011**, Atlanta, BAPS.2011.DAMOP.U5.10.

Honors and Awards

2014 Planning and Budgeting Committee (PBC) fellowship for post-doctoral researchers, the Council for Higher Education, and the Ministry of Finance, *Israel*.

2012 Qualified in a global competition among young scientists worldwide to participate in the 62nd Lindau Nobel Laureate Meeting (Physics), *Germany*.

2009 Rank 1st in the Doctoral Qualification Exam to pursue a Ph.D. degree directly, *ECNU, China*.

Teaching and Mentoring Experience

Teaching

Spring 2010 Teaching Assistant at Atomic Molecular and Photonics (graduate level). ECNU

Spring 2009 Teaching Assistant at Atomic Physics (undergraduate level). ECNU

Fall 2008 Teaching Assistant at Linear Algebra (undergraduate level). ECNU

Mentoring

2022-2023 **Aaditya Bhattacharya** (Undergraduate Student at Juniata College, PA), 2023 SPS (Society of Physics Students) Google Scholarship, Project on *Phase Amplification through the Geometric Phase in a Michelson Interferometer* Juniata College, PA

Summer 2021 **Conner Churko** (Undergraduate Student at UMD), Project on *Squeezed light and entanglement generation from four-wave mixing in hot rubidium vapor; Mechanically stabilized optical table and a three-layer temperature isolator for hot rubidium vapor.* *UMD*

2012-2013 **Jialu Yan** (Undergraduate Student at ECNU), Project on *Four-wave mixing in hot rubidium vapor.* *ECNU*
- *Later a graduate student at the University of Missouri-Columbia, USA.*

2011-2012 **Zhu Wang** (Undergraduate Student at ECNU), Project on *Diode laser building and photon-electron detection.* *ECNU*
- *Later a graduate student at the University of Wisconsin-Madison, USA.*